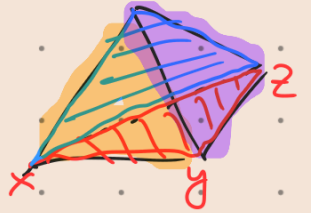


Lecture 7

Today

- (6,3)-Thm
- Roth's-Thm
- Induced matching
- Ramsey-Turán for K_4



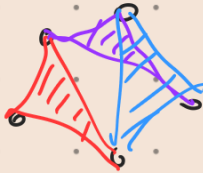
Recall 3-unif hypergraph $\mathcal{H} = (V, E)$
 $\uparrow_{x \in S} \quad \{x, y, z\}$



Open: $ex(n, K_4^{(3)}) = ?$

Def $s, t \in \mathbb{N}$, (s, t) -configuration $((s, t), \dots)$
 is a set of s vs inducing
 $\geq t$ edges

Ex $(5, 3)$ -conf.



We say $\mathcal{H}$ is (s, t) -free if $\nexists (s, t)$ -conf in $\mathcal{H}$.

I.e. $\forall S\text{-set} \subseteq V(\mathcal{H})$, we see $\leq (t-1)$ edges

Thm (Ruzsa - Szemerédi)

$\forall \epsilon, \exists n_0 = n_0(\epsilon)$ s.t.
 $\forall n \geq n_0$ the following holds

$\forall n$ -vx 3-unif $\mathcal{H}$

$(6, 3)$ -free

$\implies$

$$e(\mathcal{H}) = o(n^2)$$

($\forall 6\text{-vx} \in 2\text{ edges}$)

$$e(\mathcal{H}) \leq \epsilon n^2$$

Rank: $\exists$ lower bound construction, i.e. 3-unif n -unif
that is $(6,3)$ -free with $\frac{n^2}{e^{c \cdot \sqrt{\log n}}} (> n^{2-o(1)})$

Pf $(6,3)$ -thm. Suppose not true, i.e.

$\exists c > 0$ s.t. ∞ -many n s.t. $\exists n$ -unif $(6,3)$ -free

3-unif $\mathcal{H}$ with $e(\mathcal{H}) > cn^2$.

By zooming into a denser subgraph if necessary.

We may assume $\forall \mathcal{H}' \subseteq \mathcal{H}$, $d(\mathcal{H}) = \frac{3 \cdot e(\mathcal{H})}{n} \geq d(\mathcal{H}')$

• Maximality $\Rightarrow$ linearity  $\leftarrow$ no such pair

all such pairs are by themselves components. $\Rightarrow$ deleting such components increases ave. deg.

$$d(\mathcal{H}') = \frac{3(e(\mathcal{H}) - 2)}{n - 4} > d(\mathcal{H})$$

• $\mathcal{H}$ linear $\Rightarrow e(\mathcal{H}) = O(n^2)$ ($\leq n^2 / \frac{1}{8} \log n$)

• G = shadow graph of $\mathcal{H}$.

In G , a Δ is $\mathcal{H}$ - Δ

if it comes from an edge of $\mathcal{H}$



Note: $\mathcal{H}$ linear

$\Rightarrow \mathcal{H}$ - Δ s in G are pairwise edge-disjoint.

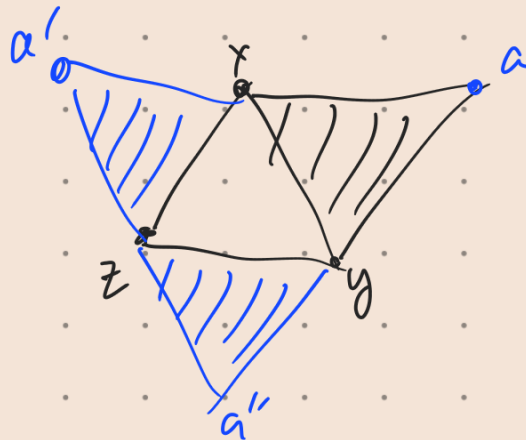


Real Δ -removal lemma: if $\exists c'n^2$ edge-disjoint Δ s
 then G has $\geq c'n^3$ Δ s.

$$\# \mathcal{H}\text{-}\Delta\text{s in } G = e(\mathcal{H}) \geq cn^2$$

$$\Rightarrow \# \Delta\text{s in } G \geq c'n^3 \underset{\uparrow \text{large}}{>} n^2 \left(\frac{1}{8} \epsilon o(1) \right) \geq e(\mathcal{H})$$

So G contains a non- $\mathcal{H}$ - Δ



$\Rightarrow (6,3)$ in $\mathcal{H} \Rightarrow$

Roth's thm

$((6,3)\text{-thm} \rightsquigarrow \text{Roth's thm})$

Roth's $\forall \delta > 0, \exists n_0 = n_0(\delta)$ s.t. $\forall n \geq n_0$, T.F.H.

$\forall$ set $S \subseteq [n] = \{1, 2, \dots, n\}$ with $|S| \geq \delta n$

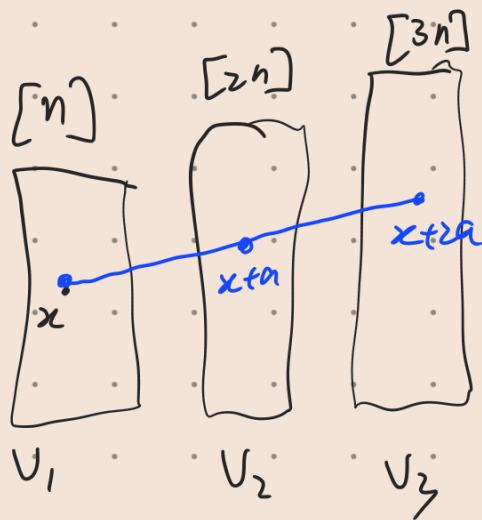
$\Rightarrow S$ contains a 3-term arithmetic progression (3AP)

$(a, a+d, a+2d)$

Pf $((6,3)\text{-thm} \Rightarrow \text{Roth's thm})$

Suppose $\exists$ 3AP-free $A \subseteq [n]$, $|A| \geq \delta n$

We shall $\xrightarrow[|A| \geq \delta n]{\text{3AP-free } A}$ 3-unif $\mathcal{H}$
 n -ux



(6,3)-free $e(\mathcal{H}) = \Omega(n^2)$
 (6,3)-thm. $\Leftarrow$

$E(\mathcal{H})$:

$\forall x \in [n], \forall a \in A$

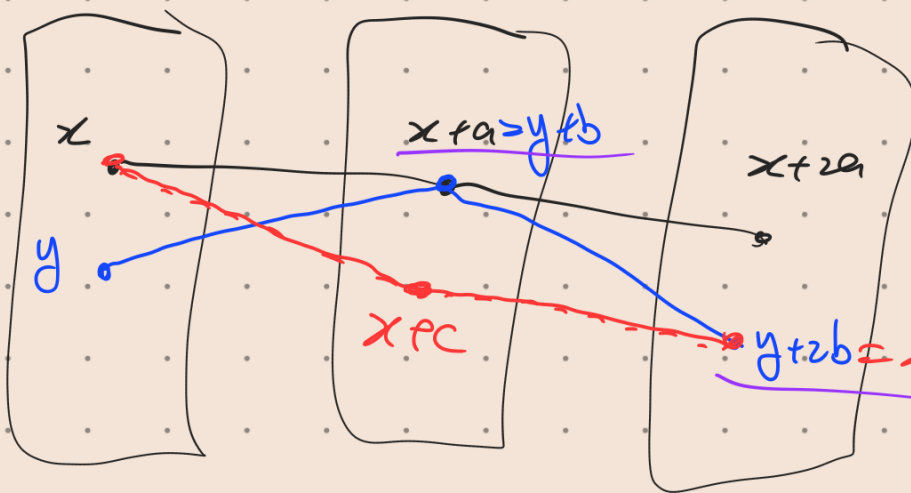
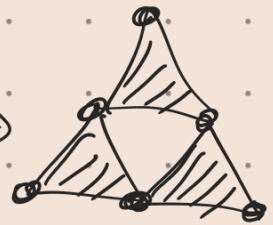
$\rightsquigarrow$ hyperedge $(x, x+a, x+2a)$

$$e(\mathcal{H}) = n |A| \geq \delta n^2 = \Omega(n^2)$$

$\Rightarrow \mathcal{H}$ contains a (6,3)-conf.

Firstly, $\mathcal{H}$ is linear. (Ex.) $\leftarrow$

so (6,3) looks like $\Rightarrow$



$a, b, c \in A$

WTS a, b, c
 form 3AP
 to get $\Leftarrow$

$$x+a = y+b \quad (1)$$

$$x+2c = y+2b \quad (2)$$

$$(2) - (1) \Rightarrow 2c - a = 2b - b = b$$

$$\Rightarrow a + b = 2c \Rightarrow \text{3AP} \quad \Leftarrow \Rightarrow$$

Rmk: In the above pf $\mathcal{H}$ (6,3)-free $\Rightarrow A$ 3AP-free

Behrend's constr: $S \subseteq [n]$ 3AP-free with

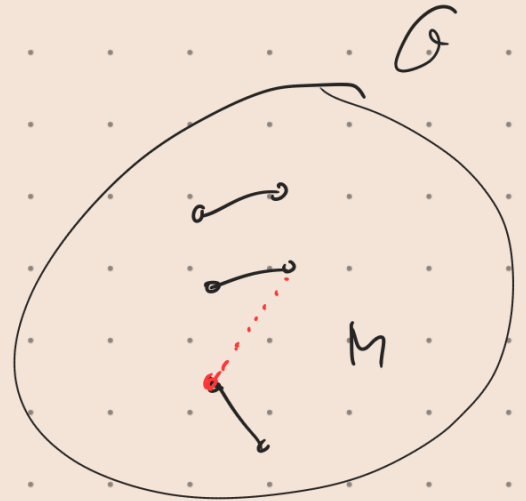
$$|S| \geq \frac{n}{e^{\sqrt{\log n}}}$$

$$\leadsto e(\mathcal{H}) \text{ (6,3)-free} \quad e(\mathcal{H}) \geq n \cdot |S| = \frac{n^2}{e^{\sqrt{\log n}}}$$

Induced matching

Def M , a matching in G , is an induced matching if

$$E(G[V(M)]) = E(M)$$



Thm 3.1 (Induced matching thm)

n -vx G

$$E(G) = \bigcup_{i=1}^n M_i,$$



$$e(G) = o(n^2)$$

where each M_i is an induced matching

Note

Thm 3.2

(6,3)-thm

Induced matching thm

HW

Roth's-thm

Pf (Thm 3.1) Suppose to the contrary that $\exists$ n -vx G
 that is $\left\{ \begin{array}{l} \text{a union of } n \text{ induced matchings.} \\ e(G) \geq cn^2 \end{array} \right.$

$G \xrightarrow[\substack{\varepsilon = \frac{c}{10}, m = \frac{1}{\varepsilon}}]{\text{reg. Lem}} R = \overset{\text{reduced}}{R}(\varepsilon, 2\varepsilon)$



Standard cleaning $G \longrightarrow G_R$

$$e(G_R) \geq e(G) - 3\varepsilon n^2 \geq cn^2/2$$

As $E(G_R) = \bigcup_{i=1}^n M_i$

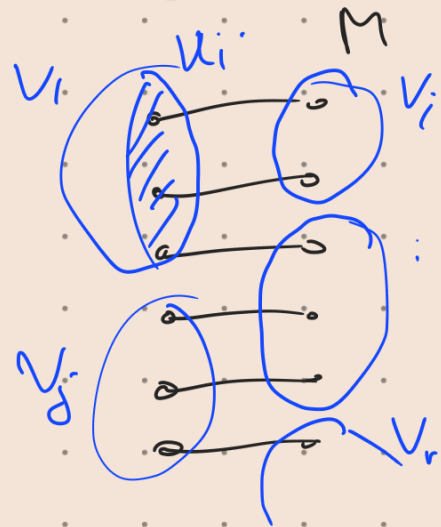
$\Rightarrow \exists$ matching $|M| \geq \frac{e(G_R)}{n} \geq cn/2$

$|V(M)| \geq cn$

Def $\forall i \in [r] = V(R)$,

$U_i = V_i \cap V(M)$

$U = \bigcup \{U_i : |U_i| \geq \varepsilon |V_i|\}$



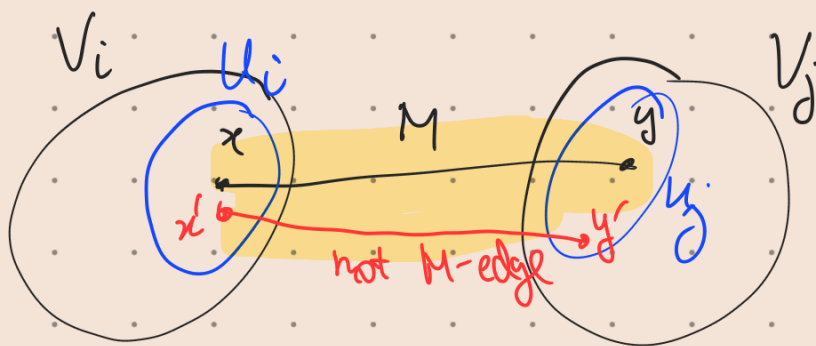
Going $\bigcup_{i \in [r]} U_i = V(M) \rightsquigarrow U$ lose $\leq \varepsilon n$ vxs

$\Rightarrow |U| \geq \frac{9}{10} cn > |M|$

$\Rightarrow U$ spans an edge in M

$\underline{xy} \rightsquigarrow \in E(G_R)$





(U_i, U_j) is ε -reg &
 $d(U_i, U_j) \geq 2\varepsilon$

$$\begin{aligned} |U_i| &\geq \varepsilon |V_i| & \varepsilon\text{-reg} \\ |U_j| &\geq \varepsilon |V_j| & \Rightarrow d(U_i, U_j) \geq d(V_i, V_j) - \varepsilon \geq \varepsilon \end{aligned}$$

$$e(U_i, U_j) \geq \varepsilon |U_i| |U_j| > |U_i| \Rightarrow \exists \text{ edge } x'y' \text{ in } [U_i, U_j]$$

& $x'y' \notin M, x', y' \in V(M)$

$\Rightarrow M$ not induced matching $\Rightarrow$

Open (73 Brown - Erdős - Gs) Let $s \in \mathbb{N}$.

$$\begin{aligned} \mathcal{H} \quad & n\text{-unif} \\ & 3\text{-unif} \\ & (s+3, s)\text{-free} \end{aligned} \Rightarrow e(\mathcal{H}) = o(n^2)$$

Rmk. (7,4) open?

Ramsey - Turán for K_4

$$\text{Turán: } G \begin{cases} K_{r+1}\text{-free} \\ n\text{-unif} \end{cases} \Rightarrow e(G) \leq$$



Def Ramsey-Turán number for H is

$$RT(n, H, m) = \max \left\{ e(G) : |G| = n, \alpha(G) \leq m, G \text{ is } H\text{-free} \right\}$$

• Classical setting $m = o(n)$.

Def: $P(H) = \lim_{\delta \rightarrow 0} \lim_{n \rightarrow \infty} \frac{RT(n, H, \delta n)}{\binom{n}{2}}$

Ramsey-Turán density of H .

$$RT(n, H, o(n)) = P(H) \cdot \binom{n}{2} + o(n^2)$$

Ex $RT(n, K_3, o(n)) = o(n^2)$

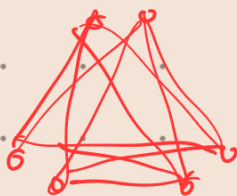
Thm (Szemerédi 73)

$$RT(n, K_4, o(n)) \leq \frac{n^2}{8} + o(n^2)$$

Open: Erdős-Simon-Stone for $RT(n, H, o(n))$?

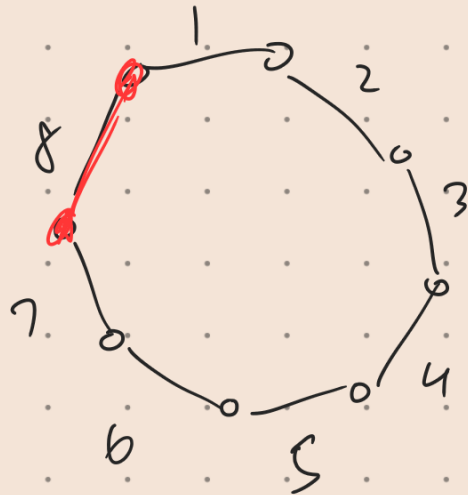
Open (Simplest case)

Is $RT(n, K_{2,2,2}, o(n)) = o(n^2)$?



$G = \text{union of } n \text{ induced matchings}$
 $n = 4x$

$$e(G) = O(n^2)$$

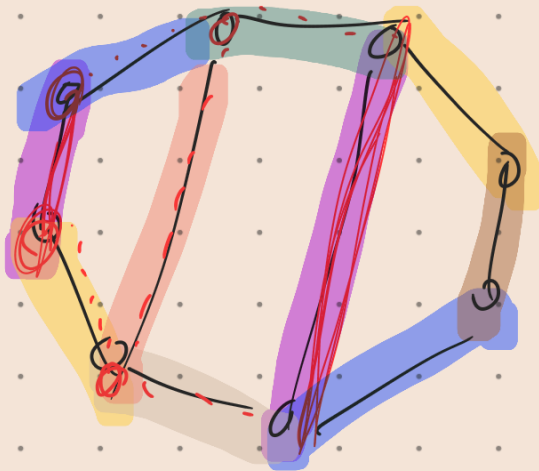


$$n = 8$$

$M_i = \text{edge } i$

$$n = 8$$

$$e(G) = 10$$



7 colours

